

Congruent or Similar? Evidence from Transformations

Using side lengths and slopes as evidence that a translation, reflection, rotation, or dilation maps one figure onto another

Directions. Saying two figures “look the same” is not evidence. On this sheet you show it with numbers: compute lengths with the distance formula and directions with slope, then say what those numbers prove.

- **Distance:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. **Slope:** $m = \frac{y_2 - y_1}{x_2 - x_1}$ (undefined when the segment is vertical).
- **Congruent** means a sequence of *rigid motions* — translation, reflection, rotation — maps one figure exactly onto the other. Every corresponding length stays the same.
- **Similar** means a dilation may be used as well. Corresponding lengths are then all multiplied by the same scale factor k .
- **Slopes tell you which motion.** A translation keeps every slope unchanged. A half turn does too. A reflection across $y = x$ swaps a slope for its reciprocal. A vertical segment has no slope at all.

Round nothing: every answer on this sheet comes out exact. Show the substitution into the formula, not just the result.

1. Triangle ABC has vertices $A(1, 1)$, $B(5, 1)$, $C(5, 4)$. It is translated to triangle $A'B'C'$ with $A'(4, 3)$, $B'(8, 3)$, $C'(8, 6)$.

Find AB and $A'B'$, and state what the two lengths show.

Answer: _____ units

2. Segment DE has endpoints $D(2, 3)$ and $E(6, 6)$. It is reflected across the x -axis to $D'(2, -3)$ and $E'(6, -6)$.

Find DE and $D'E'$, and state what the two lengths show.

Answer: _____ units

3. Segment FG has endpoints $F(0, 0)$ and $G(3, 4)$. A dilation centred at the origin sends it to $F'(0, 0)$ and $G'(6, 8)$.

Find FG and $F'G'$. Are the segments congruent? What is the scale factor?

Answer: _____ units

4. Segment HJ has endpoints $H(1, 2)$ and $J(4, 6)$. A rotation about the origin sends it to $H'(-2, 1)$ and $J'(-6, 4)$.

Find HJ and $H'J'$. Does this rotation preserve length?

Answer: _____ units

5. Segment KL has endpoints $K(-2, 1)$ and $L(2, 9)$. Its image under a translation is $K'(1, -1)$ and $L'(5, 7)$.

Find the slope of KL and the slope of $K'L'$. What do the two slopes tell you about the two segments?

Answer: _____

6. Triangle MNP has vertices $M(0,0)$, $N(4,0)$, $P(0,3)$. A dilation centred at the origin sends it to $M'(0,0)$, $N'(6,0)$, $P'(0,4.5)$.

Find MN , $M'N'$, NP , and $N'P'$. Then compute both ratios $\frac{M'N'}{MN}$ and $\frac{N'P'}{NP}$ and say what they show about the two triangles.

Answer: _____ units

7. Triangle QRS has vertices $Q(1,1)$, $R(4,1)$, $S(1,5)$. Its image is $Q'(1,1)$, $R'(1,4)$, $S'(5,1)$.

Find all three side lengths of each triangle: QR and $Q'R'$, RS and $R'S'$, QS and $Q'S'$. Then state the congruence criterion your six numbers establish.

Answer: _____ units

8. Segment TU has endpoints $T(2, 1)$ and $U(6, 4)$. A rotation of half a turn about the origin sends it to $T'(-2, -1)$ and $U'(-6, -4)$.

(a) Find TU and $T'U'$.

(b) Find the slope of TU and the slope of $T'U'$.

(c) Most rotations change a segment's slope. Explain, using your two slopes, why this one does not.

Answer: _____

9. Ana says: "Triangle ABC with $A(0, 0)$, $B(4, 0)$, $C(0, 3)$ and triangle $A'B'C'$ with $A'(0, 0)$, $B'(8, 0)$, $C'(0, 6)$ are congruent, because both are right triangles with a vertex at the origin."

(a) Find AB and $A'B'$.

(b) Use those two lengths to explain in one sentence why Ana is wrong.

(c) Name the relationship that *does* hold between the two triangles, and give the transformation that produces it.

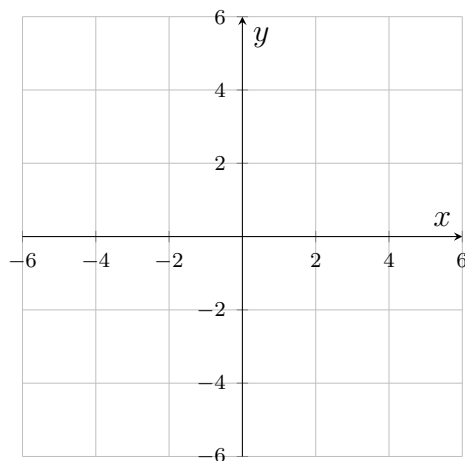
Answer: _____

10. Triangle VWX has vertices $V(1, 1)$, $W(4, 1)$, $X(4, 3)$. Its image is $V'(-1, 1)$, $W'(-4, 1)$, $X'(-4, 3)$. Plot both triangles on the grid if it helps.

(a) Find VW and $V'W'$.

(b) Find the slope of WX and the slope of $W'X'$.

(c) A single rigid motion maps VWX onto $V'W'X'$. Name it, and justify your answer using the numbers from parts (a) and (b).



Answer: _____

11. Triangle ABC has vertices $A(1,1)$, $B(5,1)$, $C(1,4)$. Triangle DEF has vertices $D(2,3)$, $E(10,3)$, $F(2,9)$.

Find all six side lengths — AB and DE , AC and DF , BC and EF — then compute the three ratios of corresponding sides. Are the triangles similar? If so, give the scale factor.

Answer: _____ units

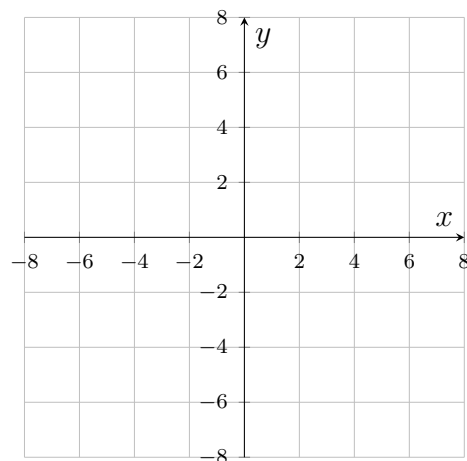
12. Synthesis challenge. Triangle JKL has vertices $J(-4, 2)$, $K(-1, 2)$, $L(-1, 6)$. Its image is $J'(2, -4)$, $K'(2, -1)$, $L'(6, -1)$.

(a) Find JK and $J'K'$.

(b) Find the slope of JL and the slope of $J'L'$.

(c) A single rigid motion maps JKL onto $J'K'L'$. Name it and justify it from your evidence.

(d) Explain why equal side lengths on their own would not have been enough to say *which* rigid motion was used.



Answer: _____